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Analysis of the impact of China Healthcare Security diagnosis-related groups payment reform on the hospitalization costs for uvulopalatopharyngoplasty in obstructive sleep apnea patients at Hospital C

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Abstract

Background As part of China's medical insurance-related grouping payment reform, public hospitals, which are the primary providers of medical insurance services, face remarkable pressure to control costs. This study examined uvulopalatopharyngoplasty (UPPP) surgery for patients with obstructive sleep apnea (OSA) at Hospital C to identify factors influencing inpatient costs and assess the impact of payment reform on treatment practices.

Methods Using data from 251 OSA patients undergoing UPPP at Hospital C, the study evaluated changes in cost reduction, cost structure, patient disease severity, and medical quality before and after the payment reform.

Results Inpatient costs for OSA patients undergoing UPPP surgery significantly decreased following the trial implementation of CHS-DRG, with notable reductions in medication and consumable expenses, as well as a shorter length of hospitalization. The CHS-DRG reform did not affect illness severity, patient safety during hospitalization, or long-term surgical outcomes for OSA patients.

Conclusion After the CHS-DRG reform, Hospital C proactively adjusted clinical pathways to manage costs and improve inpatient turnover efficiency without resorting to cost-shifting, patient refusal, or staged surgeries. Despite the financial pressures, hospitals should aim to establish a win-win mechanism between hospitals, physicians, and patients, enhance internal management, and improve financial performance. Additionally, medical security authorities should explore more precise DRG payments and value-based payment models to promote high-quality hospital development rather than impose constraints.

Study approval This study was approved by the Ethics Committee of Beijing Tsinghua Changgung Hospital (Approval No. 23418-0-02).

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Keywords Diagnosis-related groups, Obstructive sleep apnea, Uvulopalatopharyngoplasty, Hospital care, Hospitalization costs

Background

In China, public hospitals constitute the core of the healthcare system, with the proportion of total health expenditure directed to hospitals rising from 61.73% in 2015 to 63.55% in 2019 [1, 2]. As a result, hospitals have become increasingly central to controlling the growth of overall health expenditures. Internationally, Diagnosis-Related Groups (DRG) payment systems have been studied extensively, particularly in countries like the United States, Germany, and Australia [3]. These studies have demonstrated that DRG-based payments can effectively reduce hospitalization costs, while may also lead to unintended consequences, such as reduced care quality, service compression, and selective admission of patients [4–6]. In these countries, the reform has been linked to changes in hospital management practices and patient outcomes, providing useful lessons for the Chinese healthcare system [6, 7]. In China, the DRG payment system has been a recent concentration of healthcare reform, while studies are still limited, especially in terms of its impact on specialized, technically demanding procedures. Some emerging studies have begun to analyze how this reform affects the overall cost structure, quality of care, and financial sustainability of hospitals in China [8]. However, these studies mainly concentrated on general or high-volume procedures, leaving a gap in understanding how the reform influences more complex and cost-intensive surgeries.

On the basis of domestic and international experience, hospitals operating under the DRG payment system may face unintended outcomes, such as service compression, upcoding to achieve higher reimbursement, and cost shifting [8, 9]. In China's healthcare system, if medical institutions prioritize treating patient groups and conditions that generate surpluses under the payment reforms while neglecting more complex, higher-cost cases in the same DRG category, they may achieve favorable financial performance in the short term [4]. However, over time, this approach can erode the institution's technological edge and damage its reputation in specialized care, ultimately undermining sustainable operations [4, 7].

The existing literature on the DRG payment reform has scarcely addressed conditions that, despite being technically demanding and having a strong reputation, result in financial losses for medical institutions. Drawing on evidence-based medicine principles and domestic practices, such as clinical guidelines and pathways, the present study hypothesized that after the trial run of the CHS-DRG reform, hospitalization costs could increase compared with pre-reform levels, and treatment outcomes, such as disease severity and care quality, may worsen. This hypothesis was examined through a case study of uvulopalatopharyngoplasty (UPPP) for obstructive sleep apnea (OSA) at Hospital C. By concentrating on UPPP, this study would contribute to the literature by examining the financial and clinical implications of the DRG system on a technically complex surgery that is traditionally less profitable for hospitals, thereby filling an important gap in the current research.

Methods

Study design and data source

According to the OSA guideline [10], treatment options for OSA patients include non-invasive positive pressure ventilation therapy, nasal surgery, tonsillectomy, adenoidectomy, UPPP, and maxillomandibular advancement surgery. UPPP is mainly applicable to adult patients with OSA caused by obstruction in the oropharynx and nasopharynx. It can improve the sleep breathing condition of patients who cannot tolerate positive airway pressure therapy or have not responded to treatment with oral appliances [10, 11]. UPPP is a complex procedure among OSA treatments, consuming more resources than the median level in its DRG category. At Hospital C, the average cost per UPPP case is 26,603.80 RMB, which is significantly higher than the payment standards under Beijing's CHS-DRG system, such as DB35 (cleft lip/palate repair without complications) at 15,598 RMB and DB33 (cleft lip/palate repair with complications) at 20,781 RMB [12]. Moreover, as the Director of the Sleep Medicine Professional Committee of the Chinese Medical Association, Hospital C plays a remarkable role in formulating clinical practice guidelines for OSA. UPPP procedures at this hospital strictly follow clinical pathways based on these guidelines. The long-term effectiveness rate of UPPP surgery (defined as an AHI < 20 and a 50% improvement in preoperative AHI through polysomnography six months postoperatively [10]) is 64%, exceeding the conventional reported efficacy range of 40–50% [11]. Therefore, it is a typical case of a procedure characterized by high technical expertise and a strong reputation, while associated with financial losses, making it particularly representative in the context of China's healthcare system. The cost data for this study were obtained from the hospital's Electronic Health Record (EHR) system, integrated with the Hospital Information System (HIS).

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This system captured the detailed cost components, including medication, consumables, laboratory tests, and medical services, which were recorded in real-time as charges were made during the patient's stay. Each item was categorized and coded according to national healthcare cost standards. The costs were monitored using the hospital's financial and cost-control systems, ensuring that all charges were consistent with the patient's treatment and resource usage. Cost standardization across patients was achieved through adherence to the hospital's internal accounting policies, aligning with the DRG payment categories defined by the Chinese Healthcare Security system. The DRG system provides predetermined cost thresholds based on historical averages for specific treatments. Any deviations from these standards (i.e., outliers) were identified through routine cost audits and reviewed by the hospital's finance and audit departments to ensure accuracy and consistency. Furthermore, the costs of medication, consumables, and services were adjusted for inflation and changes in the pricing policies that occurred during the study period, ensuring that the data were comparable across different timeframes in the study.

A total of 251 adult patients with OSA undergoing UPPP surgery from the Otolaryngology-Head and Neck Surgery Department of C Hospital from January 2018 to June 2022 were included. The data, which were obtained from the HIS of C Hospital, included the patient's gender, age, and medical insurance type, along with clinical details like primary diagnosis, preoperative and total length of hospitalization, itemized cost data, discharge status, and incidence of unplanned return to the operating room. The medical records specifically included the patient's height, weight, apnea-hypopnea index (AHI), blood pressure, blood glucose, and minimum oxygen saturation (SPO_2). Any complaints or disputes related to these patients were managed by the hospital's administrative department. The data extracted included body mass index (BMI), Charlson comorbidity index (CCI), costs related to medication, consumables, laboratory tests, and medical services, as well as mortality rates, 48-h unplanned return-to-operating-room rates, and the number of complaints or disputes associated with these OSA patients. Additionally, the long-term efficacy of the surgery was assessed. Detailed definitions and assignments for each variable in the study are presented in Table 1.

In addition, according to clinical guidelines and experts' opinions, it was found that a history of hypertension, diabetes, heart disease and thyroid disease may impact medical costs. However, according to the clinical pathway guidelines of C Hospital, since the abovementioned diseases fall outside the scope of otorhinolaryngology, patients are required to complete symptomatic treatment for these conditions in the outpatient department before being admitted for surgery. As a result, no

Table 1 Variable names and definitions

Variable Type	Variable Name	Definition and values
Independent variable	Time	0 = Before the trial run of CHS-DRG reform; 1 = After the trial run of CHS-DRG reform
Demographic characteristics Control variables	Gender	1 = Male; 2 = Female
	Age	Continuous variables
	BMI	Continuous variables
	Type of medical insurance	0 = other medical insurance; 1 = Beijing medical insurance
Disease Characteristics Control variables	AHI[a]	1 = 5 < AHI ≤ 15; 2 = 15 < AHI ≤ 30; 3 = AHI > 30
	Minimum SPO_2[a]	1 = 85%≤SPO_2 < 90%; 2 = 80%≤SPO_2 < 85%; 3 = SPO_2 < 80%
	Nose Surgery[b]	0 = none; 1 = yes
	history of gastroesophageal reflux disease	0 = none; 1 = yes
	CCI	0 = 0 points; 1 = 1 point; 2 = 2 points and above
	Pre-operative hospital days	Continuous variables
	Total number of hospital days	Continuous variables
Dependent variable	Total cost of hospitalization	Continuous variables
	Cost of medicines	Continuous variables
	Cost of consumables	Continuous variables
	Examination and laboratory fees	Continuous variables

[a] In the guideline classification, AHI ≧ 30 and Minimum SPO_2 < 80% are both criteria for severe OSA [11]

[b] Nasal conditions, such as a deviated septum, enlarged nasal turbinates, allergic rhinitis, and chronic rhinitis are common complications in OSA patients undergoing surgery. In these cases, medical institutions have two options: either require the patient to be hospitalized again to treat the nasal issues separately, or address them concurrently with the OSA surgery

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significant differences were identified in the statistical analysis, and these diseases were therefore not discussed in this study.

Statistical analysis

The statistical analysis methods employed included descriptive statistics, univariate analysis, multivariate linear regression, and multivariate logistic regression. Firstly, descriptive statistics and univariate analysis were performed on all control, independent, and dependent variables. Next, multivariate linear regression analysis and model robustness tests were conducted to assess differences in total hospitalization costs (Supplementary Material 1) and cost structures (Supplementary Material 2) before and after the trial run of the CHS-DRG reform. Finally, to determine whether there were changes in disease severity and long-term surgical efficacy, separate logistic regression analyses and model robustness tests were conducted (Supplementary Material 3). Data analysis were carried out using SPSS 25.0 software (IBM, Armonk, NY, USA).

Results

Table 2 presents the results of descriptive statistics and univariate analyses concerning the hospitalization costs of patients with OSA before and after the trial run of the CHS-DRG reform. Following the trial operation, there was a significant increase in the proportion of patients covered by Beijing medical insurance compared with those with non-Beijing medical insurance ($P=0.005$). Additionally, there was a remarkable increase in the number of patients with a history of gastroesophageal reflux disease ($P<0.001$), while the CCI exhibited a significant decrease ($P=0.044$). Furthermore, both the total number of hospitalization days ($P<0.001$) and the total hospitalization costs ($P<0.001$) declined significantly, with notable reductions observed in medication costs ($P<0.001$) and consumables costs ($P<0.001$).

Table 2 Descriptive statistics and Univariate Analysis Results of hospitalization costs of OSA patients before and after the trial run of CHS-DRG reform

Variables	Assignment	2018–2020/$N=170$ Mean ± standard deviation Frequency/Frequency	2021–2022/$N=81$ Mean ± standard deviation Frequency/Frequency	t/X^2	P
Gender	Male	149 (87.65%)	72 (88.89%)	0.080	0.777
	Female	21 (12.35%)	9 (11.11%)		
Age		38.08 ± 8.33	36.52 ± 8.07	1.405	0.161
BMI		27.47 ± 3.54	27.37 ± 3.12	0.224	0.823
Type of medical insurance	Non-Beijing	71 (41.76%)	19 (23.46%)	7.995	0.005
	Beijing	99 (58.24)	62 (76.54%)		
AHI	(5,15]	20 (11.76%)	14 (17.28%)	3.384	0.184
	(15,30]	20 (11.76%)	14 (17.28%)		
	>30	130 (76.47%)	53 (65.43)		
Minimum SPO_2	[85%,90%)	34 (20.00%)	21 (25.93%)	1.138	0.566
	[80%,85%)	33 (19.41%)	15 (18.52%)		
	< 80%	103 (60.59%)	45 (55.56%)		
Nose Surgery	None	129 (75.88%)	65 (80.25%)	0.595	0.440
	Yes	41 (24.12%)	16 (19.75%)		
History of gastroesophageal reflux disease	None	82(48.24%)	71(87.65%)	35.818	<0.001
	Yes	88(51.76%)	10(12.35%)		
CCI	0 points	131 (77.06%)	73 (90.12%)	6.243	0.044
	1 point	32 (18.82%)	7 (8.64%)		
	≥ 2 points	7 (4.12%)	1 (1.23%)		
Pre-operative hospital days		3.93 ± 2.29	3.31 ± 3.09	1.785	0.076
Total number of hospital days		11.11 ± 2.71	9.41 ± 2.84	4.584	< 0.001
Total cost of hospitalization		27596.21 ± 5474.33	24490.81 ± 5111.91	4.291	< 0.001
Cost of medicine		4444.5 ± 1288.7	2664.57 ± 865.44	12.908	< 0.001
Cost of consumables		8297.45 ± 1893.01	6948.03 ± 1561.76	5.964	< 0.001
Examination and laboratory fees		3987.02 ± 1523.13	4245.74 ± 1462.89	1.274	0.204
Medical service fees		3924.15 ± 1082.89	3745.13 ± 1208.11	1.179	0.240

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Table 3 illustrates the effects of the trial run of the CHS-DRG reform on the total hospitalization costs for patients undergoing UPPP surgery. Following the trial operation, the average cost per patient decreased by 1,006.13 RMB ($P=0.044$), with no significant impact observed from the CCI on costs. In contrast, patients with a moderate AHI experienced an increase in costs of 1,814.87 RMB ($P=0.034$) compared with those with mild AHI, while those with severe AHI faced an even greater cost increase of 2,374.98 RMB ($P=0.001$) compared with mild AHI patients. Additionally, each extra day of hospitalization contributed to an increase in average costs of 806.87 RMB ($P<0.001$).

Moreover, patients undergoing nasal surgery incurred additional costs of 5,032.18 RMB ($P<0.001$) compared with those who did not undergo nasal surgery, and further details on these costs are presented in Supplementary Material 4. Logistic regression analysis evaluating the impact of the CHS-DRG pilot run on nasal surgery indicated that the probability of undergoing nasal surgery remained unchanged (refer to Supplementary Material 4).

Patients covered by Beijing medical insurance experienced a decrease in costs of 1,680.82 RMB ($P=0.001$) compared with those with non-Beijing medical insurance. To further explore the influences of insurance type on hospitalization costs (Supplementary Material 4), it was found that while costs for Beijing Medical Insurance patients did not exhibit significant changes, those for non-Beijing Medical Insurance patients significantly decreased. Overall, the model demonstrated notable robustness, confirming that hospitalization costs for OSA patients undergoing surgery significantly decreased following the trial operation.

The influences of the trial run of CHS-DRG reform on the cost structure of OSA patients undergoing surgery are summarized in Table 4 and Supplementary Material 2. After the trial run, the costs of medicine and consumables for OSA patients undergoing surgery significantly decreased, while examination and laboratory costs exhibited a substantial increase. However, there was no significant variation in medical service costs.

To clarify the primary reasons for changes in cost components before and after the trial run, this study categorized the fee breakdown and summarized the items that fluctuated by more than $100 per case in hospitalization costs, as detailed in Supplementary Material 6. The main drivers for changes in medicine and consumables costs included the substitution of imported omeprazole and electrodes with domestic alternatives, as well as plasma scalpel heads for the prevention and treatment of gastroesophageal reflux and UPPP. Additionally, in China, there was an adjustment in the antibiotic regimen, switching from a combination of two imported antibiotics to a single domestically produced antibiotic.

Table 3 Results of multi-factor linear regression analysis of inpatient costs for OSA patients before and after the trial run of CHS-DRG reform

Variables	B	*P*	95% CI	VIF	Robustness test
Constants	6713.57	0.011	(1545.86,11881.28)		7028.12[b]
Time	−1006.13	0.044	(−1982.79,−29.46)	1.14	−909.40[a]
CCI (Ref: CCI = 0)					
CCI = 1–5 points	−584.97	0.345	(−1803,633.06)	1.065	−536.02
CCI = 6 points	0.02	0.999	(−2484.59,2484.64)	1.042	123.32
Total number of hospital days	806.87	< 0.001	(642.54,971.2)	1.205	807.49[c]
AHI (Ref: AHI = Mild)					
AHI = moderate	1814.87	0.034	(141.33,3488.41)	1.793	1416.15
AHI = Severe	2374.98	0.001	(959.01,3790.94)	2.165	2245.94[b]
Nose Surgery	5032.18	< 0.001	(3974.04,6090.32)	1.075	5068.49[c]
Type of medical insurance	−1680.62	0.001	(−2660.66,−700.57)	1.208	−1745.34[c]
BMI	187.40	0.006	(53.22,321.58)	1.134	191.15[b]
Gender	−2732.68	< 0.001	(−4186.23,−1279.14)	1.216	−3091.12[c]
Age	69.65	0.012	(15.56,123.75)	1.088	70.53[b]
Adjusted R^2	0.615				0.631

B indicates non-standardized coefficient

[a] indicates $P<0.05$

[b] indicates $P<0.01$

[c] indicates $P<0.001$, as well as in the following content

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Table 4 Results of multivariate linear regression analysis of the cost structure of OSA patients undergoing surgery before and after the trial run of CHS-DRG reform

Variables	Cost of medicine	VIF	Consumables fee	Inspection and testing fees	Medical service fee
Constants	496.59	1.14	3676.57	1807.86	455.36
Time	−1438.22***	1.065	−1027.98***	495.67**	200.11
CCI (Ref: CCI = 0 points)					
CCI = 1–5 points	−310.90	1.042	−6.75	−601.78*	258.60
CCI ≥ 6 points	−643.78	1.205	−238.68	−373.06	626.98
Total number of hospital days	189.29***	1.793	48.58	77.41*	136.68***
AHI Index (Ref: AHI Index = Mild)					
AHI = moderate	387.57	2.165	558.12	282.49	167.22
AHI = Severe	273.15	1.075	768.94**	129.48	580.45**
Nose Surgery	656.90***	1.208	2488.32***	1331.13***	221.09
Type of medical insurance	−92.98	1.134	−512.53**	−484.02*	−73.31
BMI	31.54	1.216	50.31	−8.95	35.40
Gender	−286.24	1.088	−433.82	−733.00	−346.12
Age	9.70	1.14	−6.46	26.64**	15.67*
Adjusted R^2	0.561		0.526	0.255	0.273

*indicates $P<0.05$; **indicates $P<0.01$; ***indicates $P<0.001$

To assess the impact of the CHS-DRG reform trial run on the severity of disease in patients undergoing UPPP, logistic regression analysis was conducted. As presented in Table 5, the results of the multivariate logistic regression analysis for Time-CCI, Time-AHI, and Time-minimum SPO_2 indicated that there were no significant differences in the severity of OSA in patients undergoing surgery before and after the trial run. Specifically, the CCI for the CCI ≥ 2 group compared with the CCI = 0 group indicated no significant change ($P=0.430$), nor did the AHI index ($P=0.468$ and $P=0.772$) or the lowest SPO_2 level ($P=0.587$ and 0.597). However, the CCI = 1 subgroup exhibited a notable increase after the trial run compared with the CCI = 0 subgroup, with an odds ratio of 2.599 (95% CI: 1.054–6.410). Therefore, there was no significant reduction in disease severity among OSA patients undergoing surgery following the trial run, indicating no opt-in or shirking behavior among patients.

Regarding healthcare safety during hospitalization for OSA patients undergoing surgery, the mortality rate, the rate of unplanned returns to the operating room within 48 h, and the number of complaints and disputes were all zero before and after the trial. This indicates that the safety of care for hospitalized patients remained unchanged following the CHS-DRG reform trial run.

In terms of long-term efficacy after discharge, among 251 patients, 75 had successful follow-ups, with 48 demonstrating long-term surgical efficacy. The results of the logistic regression analysis are presented in Table 6 and Supplementary Material 3. There was no significant difference in long-term surgical efficacy between patients

Table 5 Results of multivariate logistic regression analysis of the severity of disease in OSA patients before and after the trial run of CHS-DRG reform

Variables	CCI = 1–5 points	CCI = 6 points	AHI = moderate	AHI = Severe	SPO_2 = Moderate	SPO_2 = Heavy
Intercept distance	−6.191	−10.607	−2.610	−6.510	−6.226**	−7.826***
Trial run	0.955*	1.173	−0.400	0.136	0.252	0.206
Male	1.940	−0.799	0.280	1.432	1.601*	1.615**
Beijing Medical Insurance	−0.769	0.361	1.009	2.081	0.745	0.987*
Nose Surgery	−0.005	0.500	−1.193	−1.823	0.397	−0.465
BMI	0.021	0.170	0.049	0.245	0.116	0.239***
Age	0.026	0.041	0.016	0.002	0.024	0.023
Total number of hospital days	0.068	0.013	0.174	0.139	0.010	0.007

*indicates $P<0.05$; **indicates $P<0.01$; ***indicates $P<0.001$

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Table 6 Results of logistic regression analysis of long-term surgical efficacy before and after the trial run of CHS-DRG reform

Variables	B	*P*	OR	95% CI	Robustness test
Constants	−1.74	0.127	0.176		28.347
Trial run	−0.90	0.199	0.407	(0.10,1.61)	0.273
AHI Index (Ref: AHI = Mild)					
AHI = moderate	1.22	0.239	3.383	(0.44,25.76)	4.377
AHI = Severe	0.56	0.487	1.758	(0.36,8.63)	3.168
Nose Surgery	1.49	0.032	4.429	(1.13,17.32)	3.783*
Type of medical insurance	0.10	0.865	1.099	(0.37,3.28)	1.273
Gender	-	-	-	-	0.667
Age	-	-	-	-	1.006
BMI	-	-	-	-	0.827*
Adjusted R^2	0.145				0.226

B represents non-standardized coefficient

*indicates $P<0.05$; **indicates $P<0.01$; ***indicates $P<0.001$

before and after the trial ($P=0.199$). However, nasal surgery was found to significantly improve long-term outcomes for patients, with an odds ratio of 4.429 (95% CI: 1.13–17.32).

Discussion

In response to payment reform pressures, C Hospital proactively adjusted its antibiotic usage, replaced imported omeprazole with domestically produced alternatives, and opted for Chinese-made electrodes and blades rather than imported ones (Table 4 and Supplementary Material 5). The hospital also reduced hospitalization days for patients with OSA. Despite these changes, there was no significant difference in the quality of medical services during hospitalization or in long-term surgical efficacy before and after the trial run. This suggests that, under substantial cost pressures, C Hospital carefully considered clinical needs and treatment standards while integrating DRG payment criteria to enhance clinical pathway management for this patient population. It is clear that clinical pathway management is vital for improving the quality and efficiency of healthcare service delivery [13–15].

Studies have demonstrated that hospitals under the DRG payment system may face unintended consequences, such as service compression, downcoding to achieve higher billing, and cost shifting [5]. However, C Hospital did not exhibit such phenomena during the CHS-DRG reform. This could be attributed to the fact that C Hospital is still in the early stages of reform, whereas the hospitals referenced in the literature were further advanced in their implementation. During the initial phase of DRG reform, surplus DRG payments can be maintained through enhanced service efficiency, adjustments to treatment protocols, and reduction in drug costs, without entering a negative cycle of compressing medical expenditures.

Following the trial implementation of DRG at C Hospital, medical expenses for OSA patients undergoing UPPP surgery with non-Beijing medical insurance significantly decreased, while expenses for patients with Beijing medical insurance exhibited no statistically significant variation (Supplementary Material 5). This suggests that, after thorough assessment and adjustment of clinical pathways, all patients were treated according to established protocols without incurring additional costs for those with medical insurance. Patients with medical insurance from other regions also benefited from these adjustments, as their healthcare expenses notably declined. This finding contrasts with some studies indicating that cost shifting mainly exploits the time lag in reform, resulting in increased expenses for out-of-town medical insurance patients to offset overspending by local medical insurance in China's healthcare system [7]. For instance, research from the Shandong Provincial Medical Insurance Administration Center found that cases of cross-regional medical treatment in the province were primarily mild, and the expenses for these cases were significantly higher than those for local treatments [16]. Additionally, data from 15 tertiary hospitals in Guangzhou indicated that, in response to payment reform, healthcare expenses for out-of-town medical insurance patients were temporarily elevated by 9.5%, leading to a corresponding individual burden increase of approximately 9.1% [17].

Before and after the trial run of CHS-DRG reform at Hospital C, there was no significant difference in the AHI and minimum oxygen saturation among patients with OSA. This indicates that patients with severe OSA were

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not denied admission, contrasting with findings from the majority of studies in China. Previous research generally suggests that hospitals, due to concerns about controlling overspending, mainly avoid admitting severely ill patients with costly conditions. This tendency is commonly reflected in difficulties related to hospitalization and surgical procedures, as explained by game theory, agency theory, and specific hospital management practices [18]. International studies have also demonstrated that when diseases with more complications are not sufficiently compensated under DRG groupings, unintended consequences can arise. For instance, a study in Norway found that during the first three years of DRG payment reform, hospitals were more likely to prioritize milder cases for orthopedic day surgery, as indicated by significantly longer waiting time for hospitalization among more severe patients compared with their milder counterparts. Consequently, Hospital C opted to prioritize its specialty reputation over strict cost control to ensure long-term development [8]. Thus, Hospital C prioritized its specialty reputation over strict cost control to ensure long-term development.

It is noteworthy that there was no significant difference in the proportion of nasal surgery before and after the trial run in Hospital C (Supplementary Material 4). The cost of nasal surgery was 5032.18 RMB per person-time (Table 3), accounting for 18.9% of the hospitalization expenses, with a statistically significant impact ($P < 0.001$). This indicates that Hospital C remains committed to delivering comprehensive medical services based on patient needs, even under the financial pressure of medical insurance payments. However, when payment reform results in actual financial losses, hospitals may find alternative ways to offset the shortfall [19–21]. From a rational perspective, as surgery for OSA enhances Hospital C's reputation, the hospital could control costs in the DRG quota by splitting the treatment process. For instance, it could perform nasal surgery separately from UPPP, readmitting the patient for nasal surgery one month later. This strategy would raise the total medical insurance reimbursement to 34,261 RMB for two procedures. Alternatively, the hospital could continue adjusting surgical plans, potentially compromising patients' long-term outcomes in favor of short-term cost saving.

Conclusions and recommendations

This study analyzed total hospitalization costs, cost structures, disease severity, detailed cost changes, and both short- and long-term medical quality before and after the trial implementation of CHS-DRG reform for OSA at Hospital C. The findings indicated that Hospital C was not involved in practices, such as refusing patient admissions, scheduling multiple surgeries, increasing costs for out-of-town insurance patients, or reducing medical quality during the reform. However, it exhibited behaviors allowed under Beijing medical insurance regulations, such as transferring preoperative examination costs to outpatient services, adjusting treatment plans, and controlling the use of high-cost medications. Despite facing significant cost pressures, Hospital C adhered to reform principles concentrated on quality control and cost-effectiveness, balancing income and expenditure while maintaining its specialty reputation. The reform yielded positive results that benefited both patients and the medical insurance system. Nonetheless, hospitals faced immense pressures related to overspending in the treatment of OSA.

Consequently, this study proposed the following recommendations, grounded in the principles of DRG payment and incorporating concepts, such as information asymmetry, surplus retention under random mean pricing, reasonable sharing of overspending risks, strategic purchasing of complex disease services, and innovative technology.

At the hospital level, Hospital C should design an integrated mechanism for hospital-physician-patient collaboration, concentrating on improving patient health outcomes, optimizing the patient experience, and reducing costs. The hospital should maintain its strengths in UPPP consultations and treatments, avoid shifting patients or curtailing services, and adapt to health insurance payment reforms while protecting the quality of care.

Firstly, the hospital should treat the total insurance reimbursement limit as a cap and establish a co-payment mechanism within the hospital. By extracting its information system's data, Hospital C should explore the cost-influencing factors for major diseases, use these factors as a basis for a mutual aid coefficient, and develop a precise internal payment system. Secondly, the hospital should actively expand compensation channels to offset the relatively low income from local medical insurance hospitalization services, utilizing revenue from scientific research grants, the commercialization of research, out-of-town patient hospitalizations, and outpatient services. Thirdly, patient follow-up should be strengthened to provide evidence-based support for future value-based payment models.

At the healthcare insurance department level, administrators should explore multi-factor precision payments in patient groups and advance value-based payment methods. Firstly, it is essential to strengthen the foundational research on information systems, including data quality, collection, and analysis, to enable accurate identification of cost-influencing factors for various disease groups. Secondly, it is suggested to

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collaborate with the health department to continuously promote the adoption of clinical pathways. Thirdly, drawing on the experiences of DRG payment reforms in countries, such as the United States and Germany, governments should establish an additional payment mechanism that transitions from short-term to long-term. This mechanism should include a transparent application process with clear standards. Diseases that are technologically advanced, reputable, and loss-making (with sufficient supporting evidence) should eventually be added to a new or existing DRG group after undergoing temporary payment compensation [21, 22, 23].

While the findings of the present study suggested that the CHS-DRG reform played a significant role in reducing hospitalization costs for OSA patients, it is important to acknowledge that this study design could not fully exclude the influence of other external factors that might contribute to the observed changes. Although we controlled for key clinical variables, such as patient comorbidities, disease severity, and hospital stay duration in the multivariate analysis, the possibility of concurrent changes in clinical practices, technology, or broader healthcare trends influencing outcomes remains. Moreover, the absence of significant changes in disease severity and safety indicators (e.g., CCI, AHI, mortality, and return to the operating room rates) before and after the reform could strengthen the likelihood that the payment reform was a primary driver of cost reductions. However, as with all before-and-after studies, establishing a direct causal relationship between the reform and observed outcomes is challenging. Future research utilizing more rigorous methodologies, such as interrupted time-series analyses or randomized controlled trials, is required to confirm the causal effect of CHS-DRG reform on both cost and clinical outcomes.

Abbreviations

CHS-DRG	China Healthcare Security Diagnosis-Related Groups
UPPP	uvulopalatopharyngoplasty
OSA	obstructive sleep apnea
AHI	apnea-hypopnea index
SPO_2	minimum oxygen saturation
BMI	body mass index (BMI)
CCI	Charlson comorbidity index

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Supplementary Material 6.

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Authors' contributions
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Data availability
Data supporting these research findings are available in supplementary materials and further inquiries can be directed to the corresponding author.

Declarations

Ethics approval and consent to participate
The study approval was obtained from the Beijing Tsinghua Changgung Hospital Ethics Committee (Approval No. 23418-0-02). All methods were carried out in accordance with relevant guidelines and regulations.

Consent for publication
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Competing interests
The authors declare no competing interests.

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